The Law of Reflection

> SYNOPSIS
>
> Light bounces off a shiny surface, and it does so by a rule so simple it can be written in three symbols. The whole of this note is that rule, the convention it is measured in, and why that convention is the part people get wrong.

BEFORE YOU START — PREDICT

A ray of light strikes a flat mirror, making an angle of **30° with the mirror surface**. At what angle does it leave, **measured from the normal**? Write down a number before you read on. If you answered 30°, the figure below is for you.

PERFORMANCE EXPECTATIONS

1. **State** the law of reflection, and say what each angle in it is measured from.
2. **Draw** the normal at a point of incidence, and mark θ_i and θ_r correctly on a ray diagram.
3. **Convert** between an angle given from the surface and the same angle measured from the normal.
4. **Predict** the direction of a reflected ray given the direction of the incident ray.

3. Law of reflection

Light bounces off a shiny surface.

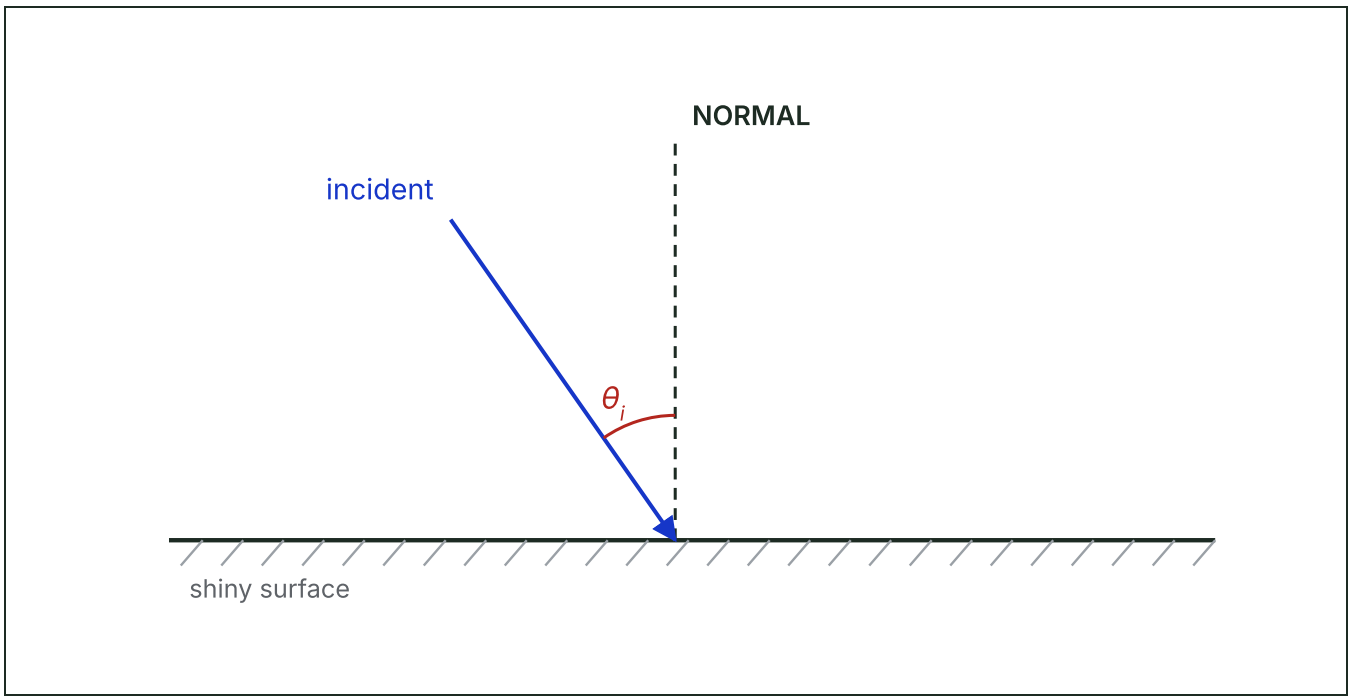


Figure 1. Both angles are measured **from the normal** — the dashed line standing at right angles to the surface — and never from the surface itself. That convention is the single most common place to go wrong, and every equation in this module assumes it.

Your job on Figure 1. The mirror, the normal and the incoming ray are drawn for you. Draw the reflected ray, mark θ_r on it, and check your angle against θ_i with a protractor rather than by eye.

$$\theta_i = \theta_r$$

The angle the light arrives at equals the angle it leaves at. Nothing in that statement depends on the colour of the light, on how bright it is, or on what the mirror is made of.

THE PUNCHLINE

$$\theta_i = \theta_r, \text{ both measured from the normal}$$

Back to the prediction: 30° from the *surface* is **60° from the normal**, so the ray leaves at 60°, not 30°. Three symbols and one convention — and the convention is the half that costs marks. Read every angle in this module from the normal, and keep doing it when the surface stops being flat and starts being a lens.

Looking ahead

A shiny surface sends the light back. A *transparent* one lets it through instead — but not along the same line it arrived on. **Note 3** is the law of refraction, and the bending that goes with it.

2. The law of reflection says the two angles are equal. Equal to what, exactly?

○ Both measured from the mirror surface. ○ Both measured from the normal to the surface. ○ The incident angle from the surface, the reflected angle from the normal. ○ Both measured from the horizontal, whatever the surface is doing.

3. A ray arrives along the normal itself, straight at the mirror. What happens?

○ It is absorbed; there is no reflected ray. ○ It reflects at 90° and travels along the surface. ○ It comes straight back the way it came, since $\theta_i = \theta_r = 0$. ○ It cannot be answered without knowing the mirror material.

4. A ray hits a mirror with $\theta_i = 38°$. Through what **total** angle is it turned, in degrees?

[ ]°

Check Clear

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